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# Bayesian Techniques in Macroeconomics

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Master in Quantitative Economics, second year

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## Chapter 3

# Maximum Likelihood Estimation

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### Handout Objectives

- Using filters to recover the sequence of forecast errors.
- Constructing the likelihood function based on the filter's output.
- Optimizing a likelihood function.
- Applying these concepts via Dynare.

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### Session roadmap

**Economic intuition.** Structural shocks are latent drivers of observed series. **Econometric intuition.** A solved model becomes a state equation. Measurement equations connect it to data, and the Kalman filter builds the likelihood. **Data.** Document every series, unit, transformation, and missing-value rule. **MATLAB/Dynare exercise.** Maximize the likelihood and diagnose identification and stochastic singularity. **Takeaway.** The model–observable mapping is part of the model.

### Acknowledgement

I thank Ghassane Benmir for providing the Kalman filter codes.

# 1. Introduction

There are several ways to parameterize a macroeconomic model. These methods can be listed as follows:

- Calibration.
- Matching Moments.
- Estimation:
  - Maximum Likelihood.
  - Bayesian Estimation.

Calibration assigns parameter values using external evidence and selected moments rather than maximizing the full likelihood. The model that is calibrated is not meant to provide a full description of some economic data, instead, it aims at describing simple economic mechanisms. This is usually addressed through the lens of simple toy models. The problem with the calibration strategy is that it is difficult to choose the right values for the parameters. The difficulty of the calibration strategy can be summarized in a critique by Pindyck [2017] of economic models of climate change as follows: "*Because the modeler has so much freedom in choosing functional forms, parameter values, and other inputs, the model can be used to obtain almost any result one desires (...)*".

In contrast, the estimation strategy relies on a positive approach: it is aimed at replicating either some specific features of the data (aka 'limited-information method'), or all the information offered by the data (aka 'full information method'). This dichotomy is referred to by econometricians as the weak versus full-information methods. Note the first prototype of a DSGE model by Kydland and Prescott [1982] was based on the weak information method in the sense that the model was calibrated to match some moments of the data. Econometricians needed more effort in research to align the model closely with the data. Taylor and Uhlig [1990] and Fair and Taylor [1990] provide some examples of estimated DSGE models using the maximum likelihood method. This handout provides an introduction to the estimation of linear models using full information methods.

# 2. Filters

Filters are statistical methods which allow us to extract relevant information from data. For a macroeconometrician, filtering consists of using the model to extract the sequence of shocks from macroeconomic time series. Note that for the same sample, there are an infinite number of sequences of shocks and models that can replicate that sample. So the extracted sequence of shocks of one model provides one of many possible representations of the data. The key aspect of the estimation is therefore to pick the set of parameters that maximizes the joint density that is the most favored by the data.

## 2.1 The Kalman filter

The Kalman filter was initially developed for engineering control problems (e.g. NASA space program). In this section, we follow Hamilton's notation and will set up the Kalman filter on general time-series model. First, the Kalman filter is linear, so this filter can only be employed for the estimation of linear models. Since many DSGE models are simulated using a linear approximation, the Kalman filter has become very popular for the estimation of DSGE models.

First, note that when we try to put the model to the data, there are observed and unobserved variables. Think about an RBC model with physical capital, consumption, productivity and investment, an econometrician is able to observe consumption and investment because these are available as macroeconomic time series, while physical capital and productivity are not directly observable.[2]

The latent state vector evolves according to:

$$x_t = Fx_{t-1} + G\varepsilon_t, \text{ with } \varepsilon_t \sim \mathcal{N}(0, Q) \tag{1}$$

where $E(\varepsilon_t \varepsilon_t') = Q$ is the covariance matrix of the structural innovations. The transition matrix $F$ need not be full rank. This transition equation describes the evolution of the state vector and corresponds to the solved policy rule of the model. Regarding the set of observable variables, the equation reads as:

$$y_t = Hx_t + v_t, \text{ with } v_t \sim \mathcal{N}(0, R) \tag{2}$$

where $v_t$ is a vector of measurement errors with covariance $R$. The matrix $H$ maps the state vector into observables. These relations are called measurement equations. Substituting the transition equation into the measurement equation gives:

$$y_t = HFx_{t-1} + HG\varepsilon_t + v_t \tag{3}$$

Note that we don't allow for the stochastic disturbances $\varepsilon_t$ and $v_t$ to be correlated (but we could relax this assumption).

Back to the Kalman filter, the latter allows an econometrician to retrieve the conditional estimates of the latent states $x_t$ from the observations $y_t$. Shock estimates can then be obtained from the filtering and smoothing recursions. We assume for now that $H$, $F$, $G$, $R$ and $Q$ are known and time invariant for this presentation. We provide observations $Y$, all five matrices and an initial state distribution. The filter returns prediction errors and conditional state moments.

To distinguish the latent state from its estimate, define

$$x_{t|s} = E(x_t \mid Y_{1:s}), \qquad P_{t|s} = \text{Var}(x_t \mid Y_{1:s}).$$

[2]For physical capital, we are not sure exactly at which rate physical capital depreciates. Similarly, the measure of productivity is also subject to endless controversies (think about the Solow residual).

Prediction uses observations through $t-1$, filtering through $t$, and smoothing through the sample end $T$. Assume Gaussian innovations independent over time and independent of the initial state, with mutually independent structural and measurement innovations. The Kalman recursions then give exact conditional moments. Without Gaussianity, they give the best linear predictor under the stated moment assumptions, not generally the full conditional distribution.

The one-step prediction error and its conditional covariance are

$$S_t = y_t - Hx_{t|t-1}, \tag{4}$$
$$\Omega_t = HP_{t|t-1}H' + R. \tag{5}$$

The conditional mean minimizes mean squared prediction error. Any predictor $a$ based on past observations satisfies

$$E_{t-1}\|y_t - a\|^2 = E_{t-1}\|y_t - E_{t-1}y_t\|^2 + \|a - E_{t-1}y_t\|^2.$$

The cross term vanishes because the prediction error has conditional mean zero. For nonsingular $\Omega_t$, the gain scales the surprise by state–observation covariance relative to forecast uncertainty:

$$K_t = P_{t|t-1}H'\Omega_t^{-1}. \tag{6}$$

Updating and forecasting give the remaining four recursions:

$$x_{t|t} = x_{t|t-1} + K_tS_t, \tag{7}$$
$$P_{t|t} = P_{t|t-1} - K_tHP_{t|t-1}, \tag{8}$$
$$x_{t+1|t} = Fx_{t|t}, \tag{9}$$
$$P_{t+1|t} = FP_{t|t}F' + GQG'. \tag{10}$$

Here $P$ replaces the older shorthand $\Sigma$ for state-estimation error covariances. The arrays Ex_hat, x_hat, ESigma and Sigma store predicted mean, filtered mean, predicted covariance and filtered covariance, respectively.

Initialization is part of the likelihood. A known $x_0$ gives $x_{1|0} = Fx_0$ and $P_{1|0} = GQG'$. A stationary initialization instead solves $P = FPF' + GQG'$. The teaching script kalmanRBC.m starts with $x_{1|0} = 0$ and $P_{1|0} = I$. This uncertain initial state does not define the same conditional likelihood as inversion from known $x_0 = 0$.

## 2.2 The Inversion Filter

As an alternative to the Kalman, we can use the inversion filter of Kollmann [2013] that recovers shocks by solving the observation relation. The displayed inverse applies to the linear case. Nonlinear inversion additionally requires existence and uniqueness.

Consider a selection matrix denoted $H$ such that the set of observables is given by $y_t = Hx_t$. The matrix $H$ has $n_y$ rows and $n_x$ columns, where $n_y$ is the number of observables and $n_x$ the number of states. A selection matrix contains one unit entry in each row and zeros elsewhere. Mapping model observables $y_t$ to the data $Y_t$ implies:

$$\begin{aligned} Y_t &= y_t \\ &= Hx_t \\ &= HFx_{t-1} + HG\varepsilon_t \end{aligned}$$

We thus solve the equation in $\varepsilon_t$ such that the closed-form expression for the contemporaneous realization of shocks:

$$\varepsilon_t = (HG)^{-1}(Y_t - HFx_{t-1}) \tag{11}$$

Note that the error of forecasts, denoted $Y_t - HFx_{t-1}$, is the economic surprise that is rescaled to the model's shock size through the product with the term $(HG)^{-1}$. Therefore using this equation, we can extract the sequence of errors recursively.

To summarize, the inversion filters work by inverting the policy rule to infer the sequence of shocks and then updates the state variables as follows:

$$\begin{cases} \varepsilon_t = (HG)^{-1}(Y_t - HFx_{t-1}) \\ x_t = Fx_{t-1} + G\varepsilon_t \end{cases},$$

for $t = 1, \ldots, T$. This recursion requires a known initial state and a square, nonsingular matrix $HG$. For state variables expressed as deviations from steady state, the deterministic initial condition is $x_0 = 0$.

## 2.3 The particle filter

Particle filters allow the inference of dynamic macroeconomics models, the latter can be non-linear and/or with non-normal errors. We refer to Fernández-Villaverde and Rubio-Ramírez [2007] for a full description of the application to DSGE models.

## 2.4 A quick discussion

The Kalman filter tracks state uncertainty and, in a linear Gaussian model, provides conditional state means and covariances. Inversion can recover a unique shock sequence conditional on a known initial state, no measurement error, and a square, nonsingular matrix $HG$. Equal numbers of shocks and observables alone do not establish invertibility or eliminate uncertainty about the initial state. With more shocks than observables, the Kalman filter remains applicable even though the direct inverse does not exist. For nonlinear or non-Gaussian models, particle filters provide a numerical alternative, often at a substantially higher computational cost.

Now that we are able to get the sequence of errors that replicate some observable variable, the next step is to pick the best set of parameters that replicates the data. The maximum likelihood is the criterion for this.

# 3. Maximum likelihood

In statistics, maximum likelihood estimation (MLE) is a method of estimating the parameters of a probability distribution by maximizing a likelihood function, so that under the assumed statistical model the observed data is most probable. From a statistical standpoint, a given set of observations are a random sample from an unknown population. The goal of maximum-likelihood estimation is to make inferences about the population that is most likely to have generated the sample.

## 3.1 An illustration with a simple Gaussian case

For an independent and identically distributed Gaussian sample, consider the forecasting model:

$$x_t = \varepsilon_t \text{ with } \varepsilon_t \sim \mathcal{N}(\mu, \sigma^2).$$

The forecast error of this model reads as follows:

$$S_t = x_t - E_{t-1}\{x_t\} = \varepsilon_t - \mu.$$

Conditional likelihood associated with a realization of $x_t$ can be assessed as the likelihood assigned by its probability density function. Note that if shocks $\varepsilon_t$ are Gaussian, then the error of predictions are Gaussian, $S_t \sim \mathcal{N}(0, \sigma^2)$. By assuming normality, we simply assume the shape of our data distribution to conform to the Gaussian bell curve. As we know from statistics, the specific shape and location of our Gaussian distribution come from $\sigma$ and $\mu$ respectively. These two parameters are what define our curve, as we can see when we look at the Probability Density Function (PDF) of the Gaussian distribution:[3]

$$\begin{aligned} P(\varepsilon_t) &= \frac{1}{\sqrt{2\pi V(\varepsilon_t)}} \exp\left(-\frac{(\varepsilon_t - E(\varepsilon_t))^2}{2V(\varepsilon_t)}\right) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\varepsilon_t - \mu)^2}{2\sigma^2}\right) \end{aligned}$$

For continuous data, the probability of an exact realization is zero. The relevant object is the density $f(x_t \mid \theta)$, where $\theta = (\mu, \sigma)$ and $\sigma > 0$. Above, $P$ is historical shorthand for a density, not a point probability. A single observation has a density, even though estimating both parameters from one observation is ill-posed.

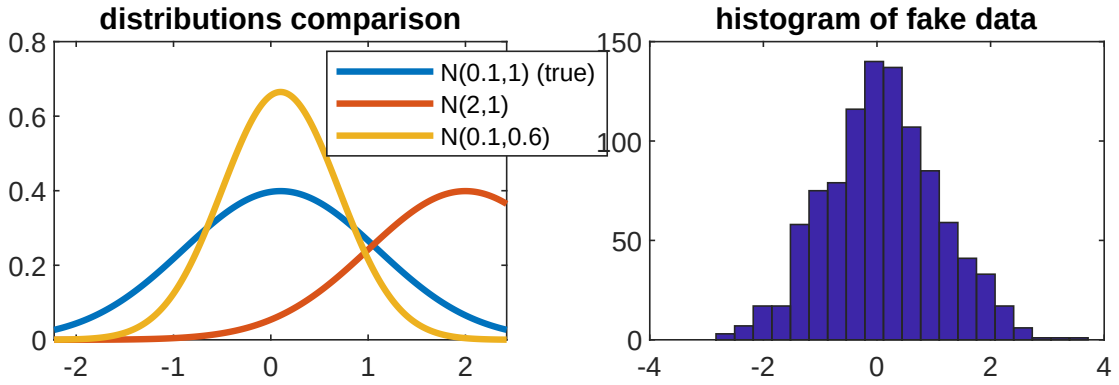
The likelihood holds the observed sample fixed and varies $\theta$. Independence in this toy example gives the product below. For a time series, the factorization instead uses conditional densities $f(y_t \mid Y_{1:t-1}, \theta)$.

$$\begin{aligned} f(x_1, x_2, ..., x_T|\theta) &= f(x_1|\theta) \cdot f(x_2|\theta) \cdot ... \cdot f(x_T|\theta) \\ &= \prod_{t=1}^{T} f(x_t|\theta) \end{aligned} \tag{12}$$

To illustrate how the log-likelihood can serve to find the best model, consider the first MATLAB box. We sample some random draws from Gaussian distribution $\varepsilon_t \sim N(0.1, 1)$, which is referred to as the true data generating process. Using the sequence of $\varepsilon_t$ that we interpret as our sample, we compare alternative distributions drawn respectively from $\theta' = \{2, 1\}$ and $\theta'' = \{0.1, 0.6\}$ .

[3] Most of this subsection is strongly inspired by https://towardsdatascience.com/maximum-likelihood-estimation-explained-normal-distribution-6207b322e47f.

MATLAB box 1 LK.m




As reported in the left panel, the true Gaussian curve in blue ($\varepsilon_t$) is strongly different from the predictions of yellow ($\theta''$) and red ($\theta'$) distributions. These theoretical curves can also be compared with the data $\varepsilon_t$ in the right panel. Therefore, the goal of the likelihood function as described in Equation 12 is to assess which candidate density on the left best fits the data on the right. By calculating the value of the likelihood, we see that $L(Y,\theta) > L(Y,\theta')$ and exceeds $L(Y,\theta'')$ in this illustration. The script keeps the direct products to expose numerical underflow, then ranks the candidates using sums of analytic log densities. Taking the log of an already underflowed product does not recover the likelihood.

Remember, the goal is to maximize this probability density term by finding the optimal $\theta$. So we are looking for the value of $\theta$ that maximizes the probability density, which is also referred to as the maximum likelihood estimator. It can be expressed as follows:

$$\hat{\theta}_{MLE} = \arg\max_{\theta} \prod_{t=1}^{T} f(x_t|\theta)$$

This problem can be easily solved using optimization routines of MATLAB. However, it is common to apply two tricks to facilitate the manipulation and the resolution of the maximum likelihood problem:

1. Apply logs to turn products into sums (it's easier to code sums than products, it's also easier to calculate gradients on sums than products).
2. Minimize the negative log-likelihood.

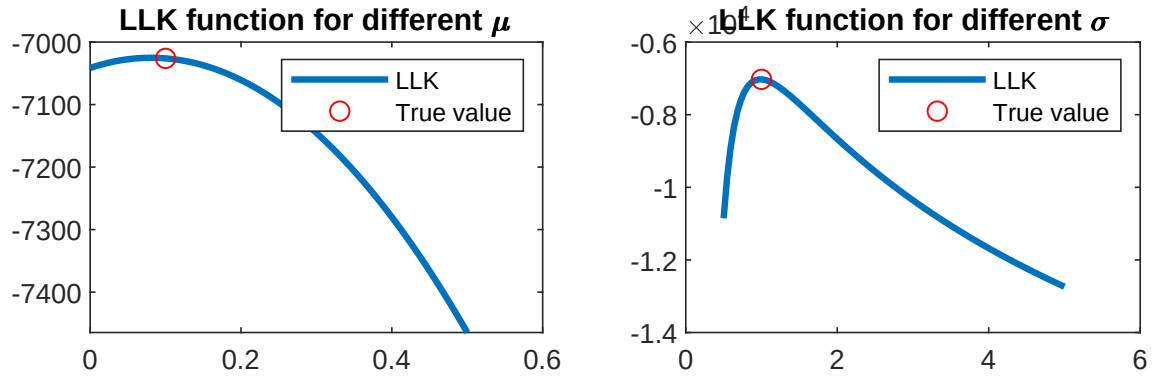
First, we take logs of the previous expression:

$$\begin{aligned}
\log(\prod_{t=1}^{T} f(x_t|\theta)) &= \log(f(x_1|\theta)) + \log(f(x_2|\theta))... \\
&= \sum_{t=1}^{T} \log(f(x_t|\theta)) \\
&= \sum_{t=1}^{T} \left[ -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(\varepsilon_t - \mu)^2}{2\sigma^2} \right] \\
&= -\frac{T}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=1}^{T}(\varepsilon_t - \mu)^2
\end{aligned}$$

Finally, in most software, it is easier to minimize than maximize, so the maximum likelihood estimator reads as follows:

$$\hat{\theta}_{MLE} = \arg\min_{\{\theta\}} \{-\log(f(x_1, x_2, ..., x_T|\theta))\} \tag{13}$$

MATLAB box 2 MLE.m




## 3.2 Maximum likelihood for DSGE models

The previous subsection describes maximum likelihood with one observed variable. With $n$ observed variables, the one-step prediction errors have conditional distribution $S_t \sim \mathcal{N}(0, \Omega_t)$ under the model. For a complete sample and nonsingular $\Omega_t$, the Gaussian log-likelihood is:

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{1}^{T}\log\left(|\Omega_t|\right) - \frac{1}{2}\sum_{1}^{T}\left(y_t - E_{t-1}\{y_t\}\right)'\Omega_t^{-1}\left(y_t - E_{t-1}\{y_t\}\right). \tag{14}$$

Kalman filter. For an estimation that uses the Kalman filter, the likelihood can be computed by picking the model-implied distribution of prediction errors in Equation 4 and Equation 5:

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{1}^{T}\log\left(|\Omega_t|\right) - \frac{1}{2}\sum_{1}^{T}S_t'\Omega_t^{-1}S_t \tag{15}$$

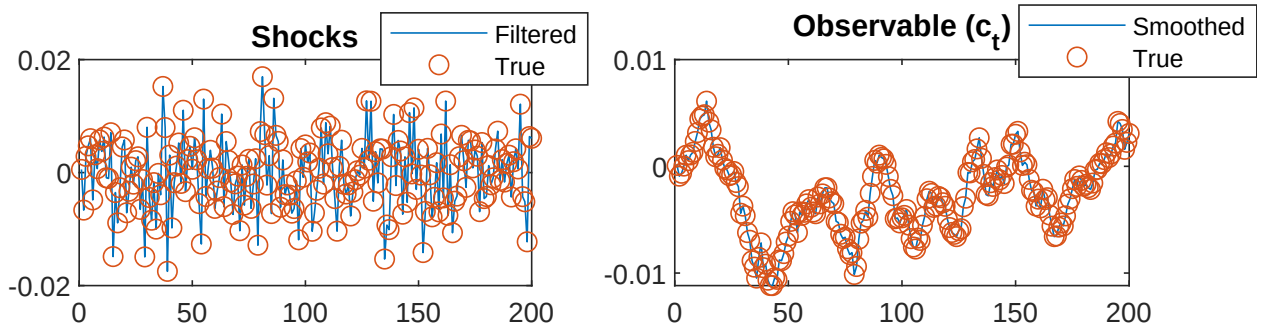
Inversion filter. Conditional on a known initial state, no measurement error, and invertible $HG$, the recovered shocks satisfy $\varepsilon_t \sim \mathcal{N}(0, Q)$. The density of the observed data also requires the change-of-variables Jacobian $|\det(HG)|^{-1}$ at each date:

$$\log L = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{1}^{T}\log\left(|Q|\right) - \frac{1}{2}\sum_{t=1}^{T}\varepsilon_t'Q^{-1}\varepsilon_t - T\log|\det(HG)|. \tag{16}$$

## 3.3 Illustration with the RBC model

Consider the RBC model with an artificial sample of $T = 200$. Recall that we need at least one shock to estimate a model with one observable. Since we only have one shock in the model, our setup only admits one observable variable. This variable will be consumption because it's one of the easy to find macroeconomic time series.

MATLAB box 3 invertRBC.m




Let us first apply the inversion filter. As seen in the MATLAB box, the inversion filter successfully retrieves the sequence of errors that generated the data. The reconstructed path of consumption, that we obtained by feeding the model with the estimated sequences of errors, exactly matches the observed path of consumption.

We can also estimate the model by maximizing the likelihood function in Equation 16 to estimate the variance of errors $Q$. The estimated standard deviation in the historical illustration is $\sqrt{\hat{Q}} = 0.0067$, close to the true standard deviation $\sqrt{Q} = 0.007$. The true variance is $Q = 0.007^2$. When only $Q$ is estimated and $F$, $G$, and $H$ are fixed, the Jacobian term is constant and does not affect the maximizing value.

Next, we can also apply the Kalman filter to perform the same exercise, we obtain similar results.

# 4. Maximum likelihood estimation with Dynare

## 4.1 Data transformation

A key problem when fitting DSGE models is to map a stationary model to non-stationary data. Macroeconomic series may contain stochastic or deterministic trends. The evidence of Nelson and Plosser [1982] warns against assuming that removing a linear trend always yields stationarity. Here, linear detrending is an explicit teaching specification whose adequacy must be checked.

There are two main components that make GDP grow steadily (as underlined by the Solow [1956] model):

1. Technological changes.
2. The population.

The first way to remove a non-stationary component from GDP is to divide it by the working population. Indeed, the growth of the population makes the GDP steadily increase at least at the same rate as the population. It is also possible to remove technological changes by rewriting the model in terms of effective labour units as in Solow's model (this step is explained in the following subsection).

First consider real GDP. Many macroeconomic series are available from DB·NOMICS. For reproducibility, use the supplied `mydb.m` in `session3/codes` and the standard MATLAB client `call_dbnomics.m`. Record the series identifier, retrieval date, sample, and raw data used in the exercise. The DBnomics guide documents the standard client. Before transformation, verify frequency, units, seasonal adjustment, dates, duplicate periods, and missing observations.

1. open https://db.nomics.world/ and verify the provider, dataset, dimensions, and series code.
2. scroll down on the main page, find and click on the OECD database, select the quarterly national accounts, and find the public series identifier (not an API credential). The historical series identifier used here is:

   OECD/QNA/USA.B1_GS1.LNBQRSA.Q

3. Create a MATLAB file called "mydb.m" and use the dbnomics client with the public series identifier to import the series into MATLAB.
4. Edit mydb.m after installing the standard DBnomics client on the MATLAB path. The import step is:

```
[my_mat,~,dates_nb] = call_dbnomics('OECD/QNA/USA.B1_GS1.LNBQRSA.Q');
```

5. The first output is a matrix whose first column contains MATLAB datenums and whose subsequent columns contain observations. The second output is a table. The third output, dates_nb, contains readable numeric dates, not datetime objects. Convert the first column to a MATLAB datetime vector T for plotting and verify quarterly continuity. Here gross_y denotes the GDP before transformation (stored in the 2nd column of the data matrix).

```
T = datetime(my_mat(:,1),'ConvertFrom','datenum');
gross_y = my_mat(:,2);
```

6. Recall that in the model, each random variable $\hat{x}_t$ is expressed in terms of detrended log deviations from the steady state:

$$\hat{x}_t \simeq \log\left(\frac{X_t}{\bar{X}_0\gamma^t}\right) \tag{17}$$

where $X_t$ is the non-linear expression of $\hat{x}_t$, $\bar{X}$ is the empirical mean (or the steady state) of $X_t$ and $\gamma^t$ is the trend component. We transform our sample in the same vein, we first express in terms of log deviations from the trend $\hat{X}_t \simeq \log\left(X_t/\gamma^t\right)$, we store the result in a variable called ln_gap in MATLAB. Then, we remove the steady state, *i.e.* the mean of the sample such that $\hat{x}_t = \hat{X}_t - \mu\left(\hat{X}_t\right)$ where $\mu(\cdot)$ is the empirical mean, and add the resulting values to the chart.

To have our data consistent with the model, (a) we need to apply logs to our data, (b) remove the trend $\gamma^t$, (c) remove the mean/steady state:

a First, we apply the log. Variable ln_y is the log of output:

```
ln_y =  log(gross_y);
```

b Second we want to remove trend. To do so, we want to know the value of the trend $\gamma$ that has to be calibrated in the model before the estimation. This parameter is the slope of the trend. Simple linear regression considers only one independent variable using the relation:

$$y_t = \beta_0 + \beta_1 t + \epsilon_t$$

where $t = [1, 2....T]$ is the time index, $y_t$ is the value of log GDP and $\epsilon_t$ is the log deviation from the trend. Here, the trend component is $\beta_1 = \log\gamma$. We perform the estimation of the trend in MATLAB through:

```
X = [ones(length(ln_y),1) (1:length(ln_y))'];
OLS= X\ln_y;
trend_y = X*OLS;
gamma = exp(OLS(2));
```

A gross trend factor of $\gamma = 1.005$ corresponds to quarterly growth of approximately 0.5%. Its estimated value depends on the sample. Variable ln_gap denotes the detrended log output. With an OLS intercept its sample mean is already zero up to numerical precision.

c Finally, center the detrended log series. This normalization is not an estimate of the structural steady state:

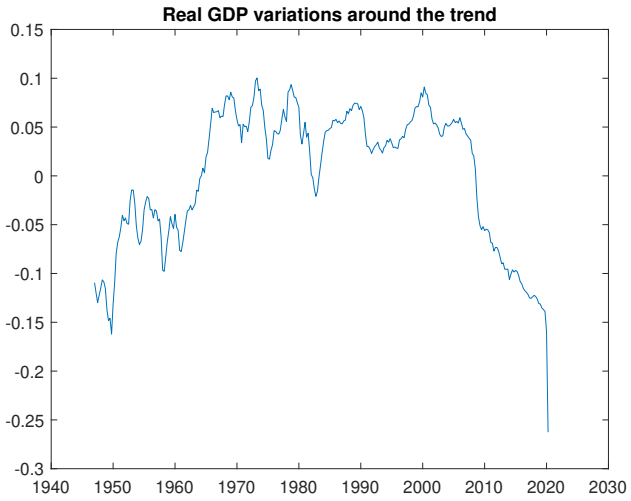



Figure 1: Log deviation of US real GDP from its fitted linear trend

```
ln_gap = ln_y-trend_y;
y = ln_gap-mean(ln_gap);
```

We can plot our result:

```
figure;
plot(T,ln_y,T,trend_y)
title('Real GDP in the downloaded sample')
figure;
plot(T,y)
title('Real GDP variations around the trend')
```

And save the transformed series of output into a matrix file, called mydata.mat, which will be read by Dynare right after for the fit exercise.

```
save mydata.mat y T trend_y OLS gamma;
```

Figure 1 plots the transformed GDP, which can be interpreted as a measure of the detrended output gap. It is thus possible to explain the gap between the real GDP and its linear trend through productivity shocks.

Important note: run mydb.m before calling Dynare so that mydata.mat exists. The variable y should be declared as an endogenous variable to be estimated in the Dynare file.

## 4.2 Syntax for log-likelihood estimation

First, let us take back the dynare file of the production economy and include an equation for output.

Second, to estimate using maximum likelihood techniques, first declare the variables of the model which are observed. This is done using the varobs VAR1 VAR2 VAR3 ; command, VARX must be in the data file a vector of an observed variable, otherwise the estimation won't start. Here, we only have output as observed:

code

```
varobs y;
```

Next, we specify the parameters to be estimated in the estimated_params; block. When writing the estimated params section, each parameter to be estimated needs to be given an initial value, and can be given bounds: the syntax to do this is parameter, initial value [, lower bound, upper bound ]. where the parts in the square brackets are optional. For the standard errors of the shocks, we use the same syntax, preceding the exogenous variable with stderr, like for specifying the shocks. The full estimated params section is:

code

```
estimated_params;
    stderr e_a, 0.007, , inf;
    rho_A, 0.95, 0, 0.999;
end;
```

Finally, we need to call estimation and specify where Dynare can find the observed series. Our observed vector is in the MATLAB file "mydata.mat". This is called with:

code

```
estimation(datafile=mydata,smoother) y c k a;
```

## 4.3 Results from the estimation exercise

Dynare optimizes the persistence parameter and shock standard deviation. The Kalman filter evaluates the likelihood, while smoothing subsequently estimates the shock sequence. The console excerpts and figures below are historical illustrations, not newly reproduced estimates for 2026–2027. The reported final value −916.4686337 is the negative log-likelihood, so the corresponding log-likelihood is positive.

**console response**

```
Improvement on iteration 10 = 0.000000031
improvement < crit termination
Final value of minus the log posterior (or likelihood):-916.468634
RESULTS FROM MAXIMUM LIKELIHOOD ESTIMATION
parameters
 Estimate s.d.  t-stat
rho 0.8362 0.0413 20.2354
standard deviation of shocks
 Estimate s.d.  t-stat
e 0.0106 0.0004 24.2567
Total computing time : 0h00m05s
```

According to the estimation output, the optimized parameters are:

$$\rho = 0.8362$$
$$sd\left(\eta_t\right) = 0.0106$$

Dynare also reports approximate standard errors and t-statistics. Under regularity conditions, $|t| > 1.96$ rejects a zero null in a two-sided 5% Wald test. This rule is not a general identification diagnostic and does not apply unchanged to variance parameters on the boundary of their support.

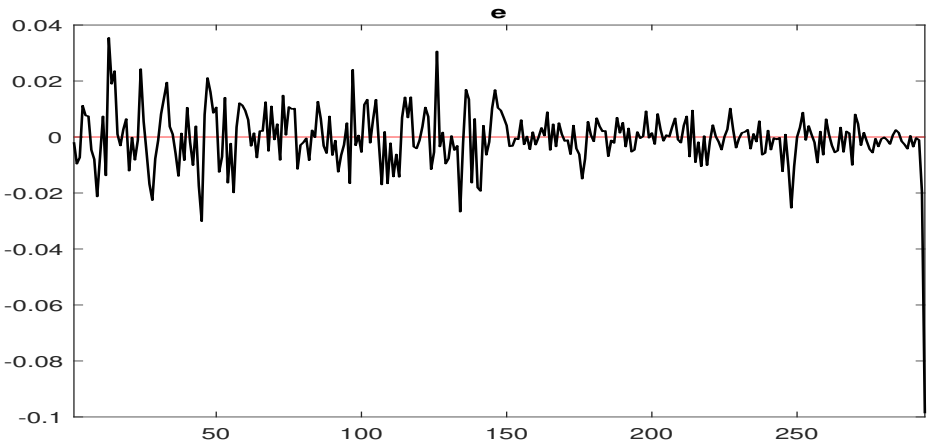



Figure 2: The sequence of estimated smoothed shocks $\{\eta_t\}_{t=1}^{T}$ over the sample period

Figure 2 plots the estimated sequence of the productivity shock that best reproduces the observed GDP over the sample period.

Finally, we can compare the observed versus the model-generated series in Figure 3. With one observable and one invertible shock, an exact in-sample fit can be mechanical. It is not independent evidence of empirical validity.

The estimation results are saved in the `oo_` structure in the workspace. Some details are given in Table 1.

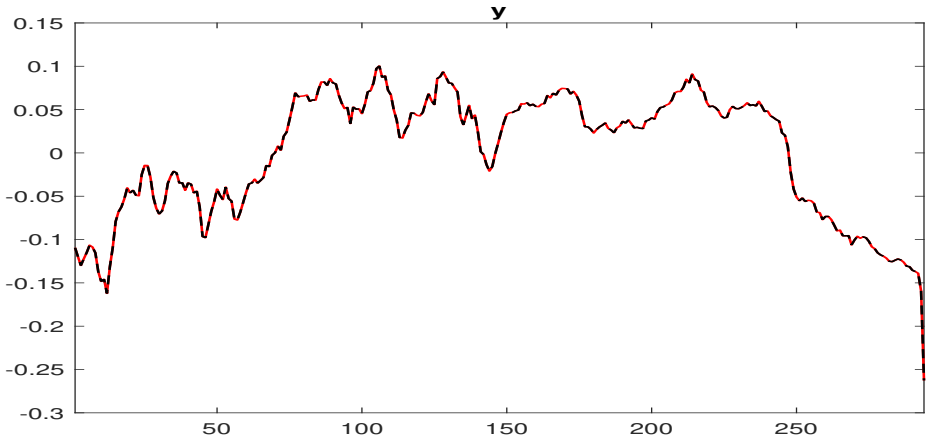



Figure 3: Observed series (black) and model-generated series (red) which use the estimated sequence of the productivity shock drawn in Figure 2

| Stored in | Description |
|---|---|
| oo_.SmoothedVariables | Estimated values of each endogenous variable for all time periods. |
| oo_.SmoothedShocks | Estimated values of each shock for all time periods. |

Table 1: MATLAB's workspace after a likelihood estimation

It is possible to reverse the detrending method in order to get estimated data back to an illustrative level path. This operation may look useless at first sight, but it allows us to plot and analyse the unobserved path of capital and productivity over the business cycles. First, we can plot detrended series:

code

```
load mydata;
figure
subplot(3,1,1);
plot(T,oo_.SmoothedVariables.y)
title('Output')
subplot(3,1,2);
plot(T,oo_.SmoothedVariables.k)
title('Capital deviations')
subplot(3,1,3);
plot(T,oo_.SmoothedVariables.a)
title('Productivity')
```

The historical paths in Figure 4 associate persistent negative output gaps after the financial crisis with lower inferred capital. These are model-dependent estimates, not separately observed quantities. Changing realized productivity volatility does not by itself establish a structural break in innovation variance or identify its cause.

To reverse the detrending method, we just need to add up the trend component and apply the exponential function $\exp(\cdot)$ to reverse the log. The productivity shock is stationary. Applying the GDP trend to

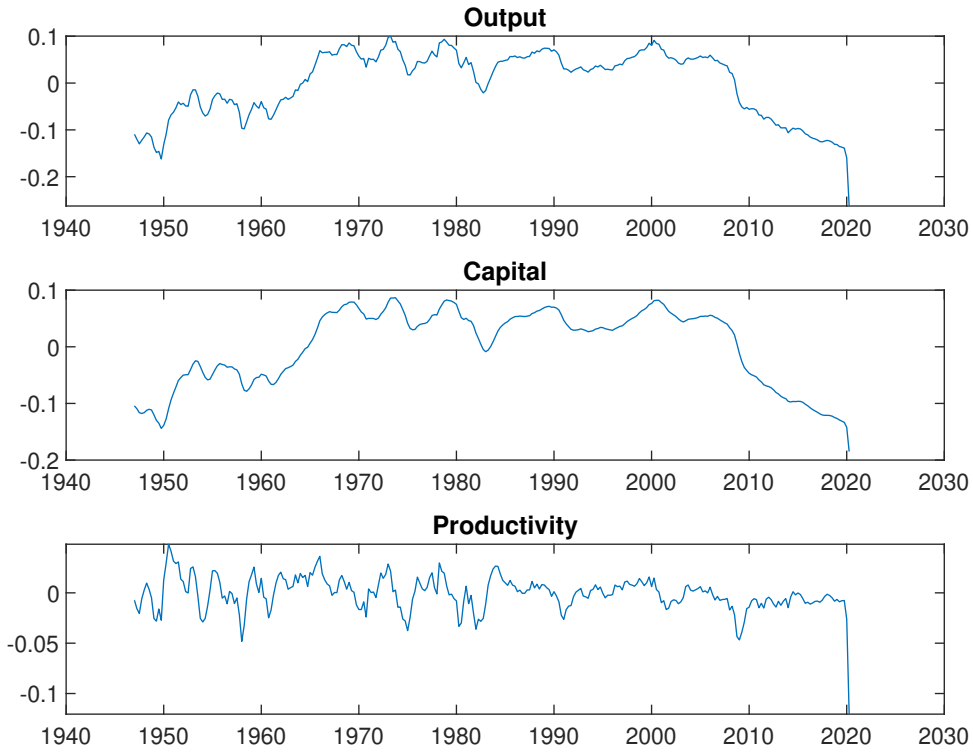



Figure 4: Model-generated output, capital and productivity (using the estimated model and sequence of shocks).

capital only produces an illustrative index. Recovering capital in economic units requires a capital-specific steady-state scale and trend.

code

```
% REVERSING
figure;
subplot(2,1,1);
plot(T,exp(trend_y+oo_.SmoothedVariables.y))
title('Output')
subplot(2,1,2);
plot(T,exp(trend_y-trend_y(1)+oo_.SmoothedVariables.k))
title('Capital index with output-trend growth')
```

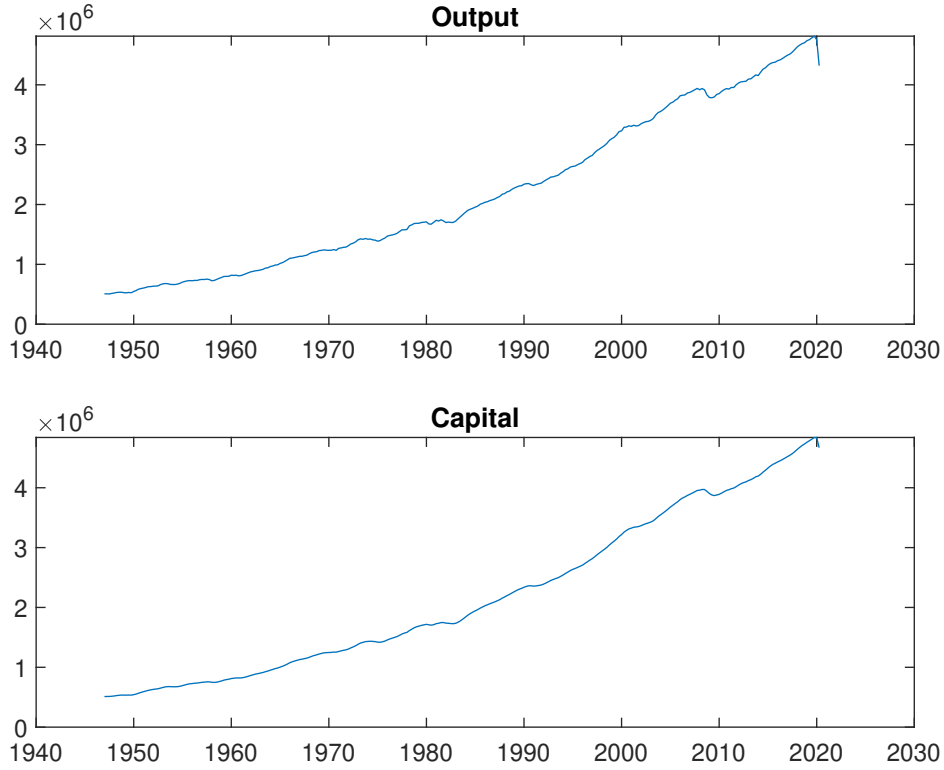



Figure 5: Historical GDP reconstruction and a capital index using the GDP trend. The capital panel is not an estimate of the capital stock in physical units

**Exercise 1: Second moment statistics and IRFs of the estimated model**

1. Compute using MATLAB's function std(·) the standard deviation of the estimated sequence of the productivity shock. Do you find the same result as Dynare?
2. Compare the standard deviation of the model with the one observed in the data. Is your model empirically relevant?
3. Why is it important to compare second moments using the same units, sample and detrending convention?
4. Plot the IRFs of the estimated model.

**Exercise 2: Estimating the model with output and consumption**

Consider now adding another time series to the model: consumption. It is thus possible to get directly from dbnomics.
Without measurement errors, having at least as many independent structural shocks as observables is a necessary dimension check, but not a sufficient rank condition. The forecast-error covariance must be nonsingular. Identification additionally requires distinct parameter values to imply distinct observable distributions. A nonzero interior estimate alone does not establish identification. To add consumption, introduce another independent shock and check its transmission to the observables.
Let's consider a public spending shock in the GDP:

$$Y_t = C_t + I_t + G_t$$

Public spending $G_t$ is financed by a tax $T_{jt}$ charged to all households populating our economy, assuming a balanced government budget over the cycles:

$$G_t = \int_0^1 T_{jt} \mathrm{d}j$$

This tax affects the budget constraint of the household:

$$C_{jt} + K_{jt} + T_{jt} = A_t K_{t-1}^{\alpha} + (1-\delta) K_{jt-1}$$

Steady-state spending is a fixed share of steady-state GDP ($\bar{G} = g^y \bar{Y}$):

$$G_t = \bar{G}\varepsilon_t^G \text{ with } \log\left(\varepsilon_t^G\right) = \rho_G \log\left(\varepsilon_{t-1}^G\right) + \eta_t^G$$

our model is now summarized by:

$$\hat{c}_t = -\beta\alpha\bar{k}^{\alpha-1}\left(\mathbb{E}_t\hat{a}_{t+1} - (1-\alpha)\,\hat{k}_t\right) + \beta\left[\alpha\bar{k}^{\alpha-1} + (1-\delta)\right]\mathbb{E}_t\hat{c}_{t+1} \tag{18}$$

$$\hat{y}_t = \hat{a}_t + \alpha\hat{k}_{t-1} \tag{19}$$

$$\bar{y}\hat{y}_t = \bar{c}\hat{c}_t + \bar{\imath}\hat{\imath}_t + g^y\bar{y}\hat{g}_t \tag{20}$$

$$\delta\hat{\imath}_t = \hat{k}_t - (1-\delta)\,\hat{k}_{t-1} \tag{21}$$

$$\hat{a}_t = \rho_A\hat{a}_{t-1} + \eta_t^A \tag{22}$$

$$\hat{g}_t = \rho_G\hat{g}_{t-1} + \eta_t^G \tag{23}$$

with $g^y = 0.2$, *i.e.* steady-state public spending is 20% of steady-state output. This ratio is not a variance decomposition. The steady state thus reads as follows:

$$\begin{aligned}
\bar{k} &= \left(\left[\frac{1}{\beta} - (1-\delta)\right]\frac{1}{\alpha}\right)^{1/(\alpha-1)} \\
\bar{\imath} &= \delta\bar{k} \\
\bar{y} &= \bar{k}^{\alpha} \\
\bar{c} &= (1-g^y)\,\bar{y} - \bar{\imath}
\end{aligned}$$

Our economy is characterized by a set of six variables $\left\{\hat{c}_t, \hat{k}_t, \hat{y}_t, \hat{\imath}_t, \hat{a}_t, \hat{g}_t\right\}$.

1. Code the new model.

2. Estimate the new model with consumption and output, estimate the variance of the spending shock as well as its AR(1) term.

3. Analyse the variance decomposition. Which shock is driving/explaining the business cycles for the US economy?

4. Is the slope of the trend different for consumption and the GDP? A common balanced-growth trend would require a consistent extension of both measurement equations.

5. Let's try to develop and compare an alternative shock which could explain the US business cycles. Include the following preference shock to the model:

$$\max_{C_{jt},K_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^\tau \varepsilon^C_{t+\tau} \log\left(C_{jt+\tau}\right)$$

   (a) How does this shock affect the Euler equation?

   (b) Estimate this new model with its AR(1) terms defined as:

$$\log\left(\varepsilon^C_t\right) = \rho_C \log\left(\varepsilon^C_{t-1}\right) + \eta^C_t$$

   (c) Which model is better given the data? (compare the model with government spending shock, preference shock and both shocks).

**Exercise 3: Estimating deep parameters**

Consider the simple model with output as its only observable. It is possible to estimate parameters such as $\beta$, $\alpha$ and $\delta$. However, these parameters affect the steady state of the model. Indeed, recall that:

$$\bar{k} = \left( \left[ \frac{1}{\beta} - (1-\delta) \right] \frac{1}{\alpha} \right)^{1/(\alpha-1)}$$
$$\bar{c} = \bar{k}^{\alpha} - \delta \bar{k}$$
$$\bar{y} = \bar{k}^{\alpha}$$

Each time $\beta$, $\alpha$ and $\delta$ are changed to maximize the log-likelihood function, the steady state should be updated accordingly. It is thus easy in Dynare to update the steady state of the model. In model(linear). 's block, add:

code

```
#Kss = ((1/beta-(1-delta))/alpha)^(1/(alpha-1));
#Css = Kss^alpha - delta*Kss;
```

At the beginning of your Dynare file, remove $Kss$ and $Css$ from the parameter block parameters ...; so that Dynare updates these parameters each iteration.
Any parameter in model's block preceded by the sign "#" is updated by Dynare each iteration of the optimization algorithm employed in the fit exercise.

1. Estimate $\delta$ and inspect its standard error, likelihood profile and sensitivity to starting values. Does the sample distinguish depreciation rates? A zero-null $t$-statistic alone does not diagnose identification.

2. Do you find a similar result for $\beta$, $\alpha$? Given the data, is your fit better with or without these parameters?

3. Compare the parameterization and information set with Smets and Wouters [2007]. Establish which parameters are estimated, calibrated or reparameterized there before comparing identification with this RBC model.

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